AP Precalculus · Unit 1 Checkpoint

Answer key, marking guide and readiness decision rules — teacher copy

Marks below are assigned by the teacher (1 mark per lettered part, 29 in total). They are not College Board scoring guidelines.

1. SOURCE MAP AND COVERAGE

Q	Official source	Lessons tested
1	2025 Q2 (all parts)	1.2 average rate of change and secant estimate; 1.3 quadratic from three points and its extreme value; 1.4 concavity
2	2024 Q2 part B	1.2 average rate of change; 1.4 concave down means secant below the graph
3	2026 Q2 part B	1.2 average rate of change; 1.4 concave up means secant above the graph
4	2025 Q1 part B	1.5 real zeros by technology; 1.6 polynomial end behavior
5–7	Original (no official source exists)	1.1 change in tandem; 1.5 complex zeros and multiplicity; 1.6 end behavior; 1.7 to 1.10 rational functions

Left out on purpose. 2024 Q1 and 2025 Q1 parts A and C and 2026 Q1 (composition, inverse functions, exponential and logarithmic models: Unit 2). Part A of 2024 Q2 and 2026 Q2 and part C (exponential and logarithmic modelling and model error: Unit 2). Question 3 (sinusoidal models) and Question 4 (equations and identities) of all three years (Unit 3 and Unit 2). Rational functions (1.7 to 1.10) and change in tandem (1.1) do not appear in any 2024–2026 free-response question, which is why Questions 5 to 7 are original.

2. SOLUTIONS AND MARKS — PART A (CALCULATOR)

1. 2025 Q2 **[6]**

Solution and marks

A(i) [1] $D(0) = 25$, so $c = 25$; $4a + 2b + c = 30$; $16a + 4b + c = 34$ (equivalent forms accepted: $a(0)^2 + b(0) + c = 25$, and so on). *Common error:* writing $D(2) = 34$ or substituting $t = 30$.

A(ii) [1] $a = -0.125$, $b = 2.75$, $c = 25$ (all three needed). By hand: $4a + 2b = 5$ and $16a + 4b = 9$ give $8a = -1$.

B(i) [1] $\dfrac{D(4) - D(0)}{4 - 0} = \dfrac{34 - 25}{4} = 2.25$ thousand plays per month. *Common error:* dividing by $34 - 25$ or using $t = 2$.

B(ii) [1] $A_{1.5} = 25 + 2.25(1.5) = 28.375$ thousand. *Common error:* evaluating $D(1.5) = 28.844$ instead of the secant estimate.

B(iii) [1] $a = -0.125 < 0$, so the graph of D is concave down on $0 < t < 4$; the secant line through $(0, 25)$ and $(4, 34)$, which is the graph of A_t, lies below the graph of D between the endpoints, so $A_t < D(t)$. Needs concave down and the secant-versus-graph position.

C [1] The vertex is at $t = -\dfrac{b}{2a} = 11$, with maximum $D(11) = 40.125$. After $t = 11$ the model decreases, but a running total of plays cannot decrease, so the domain must stop at $t = 11$ (that is, $0 \le t \le 11$).

2. 2024 Q2 PART B **[3]**

Solution and marks

B(i) [1] $\dfrac{76 - 40}{91 - 0} = \dfrac{36}{91} \approx 0.396$ thousand units per day.

B(ii) [1] $A_{50} = 40 + \dfrac{36}{91}(50) \approx 59.780$ thousand.

B(iii) [1] $G(t) = a + b\ln(t + 1)$ with $b = \dfrac{36}{\ln 92} > 0$ is increasing and concave down (its rate of change is decreasing), so the graph lies above the secant line between $t = 0$ and $t = 91$; the secant line is A_t, so $A_t < G(t)$.

3. 2026 Q2 PART B **[3]**

Solution and marks

B(i) [1] $\dfrac{14.8-27.2}{6-1}=\dfrac{-12.4}{5}=-2.48$ thousand dollars per year (negative sign required).
B(ii) [1] $A(3)=27.2+(-2.48)(3-1)=22.24$ thousand dollars.
B(iii) [1] $V(t)=ab^t$ with $0<b<1$ ($a\approx 30.721$, $b\approx 0.885$) is decreasing and concave up (the rate of change increases toward 0), so the graph lies below the secant line on $1<t<6$; the secant line is $A(t)$, so $A(t)>V(t)$. Check: $V(3)\approx 21.323<22.24$.

4. 2025 Q1 PART B **[2]**

Solution and marks

B(i) [1] $x\approx -1.233$, $x\approx 1.578$, $x\approx 5.643$ (all three needed; graph $y=g(x)$ and read the three x-intercepts, or solve numerically). *Common error:* reporting only the positive roots.
B(ii) [1] $\lim\limits_{x\to\infty} g(x)=-\infty$. The leading term $-0.167x^3$ is negative and odd. Accept "$g(x)\to-\infty$ as $x\to\infty$"; do not accept "the graph goes down" without a limit.

3. SOLUTIONS AND MARKS — PART B (NO CALCULATOR)

5. ORIGINAL **[3]**

Solution and marks

A [1] Increases: every time the input t rises, the output $d(t)$ rises ($0<12<22<30<36$).
B [1] $\dfrac{d(3)-d(1)}{3-1}=\dfrac{30-12}{2}=9$ km per hour.
C [1] Over the four hours the rates are 12, 10, 8, 6 km per hour. The rate of change is decreasing (by 2 km per hour each hour), so the graph of d is increasing and concave down. Needs the four values and a statement that the rate is decreasing leading to concave down. *Common error:* "the rate is decreasing so d is decreasing."

6. ORIGINAL **[4]**

Solution and marks

A [1] Non-real zeros of a polynomial with real coefficients occur in conjugate pairs, so $1-2i$ forces $1+2i$.
B [1] Degree 4 ($2+1+1$). $p(x)=2(x-3)^2\left(x^2-2x+5\right)$, since $(x-(1-2i))(x-(1+2i))=(x-1)^2+4=x^2-2x+5$. Accept $2(x-3)^2(x-1+2i)(x-1-2i)$ only if the factors with real coefficients are also given; the question asks for real coefficients.
C [1] $\lim\limits_{x\to\infty} p(x)=\infty$ and $\lim\limits_{x\to-\infty} p(x)=\infty$ (even degree, positive leading coefficient).
D [1] The graph touches the x-axis at $x=3$ and turns back without crossing, because the multiplicity 2 is even.

7. ORIGINAL **[8]**

Solution and marks

Write $r(x)=\dfrac{2(x-3)(x+2)}{(x-3)(x+3)}$.
A(i) [1] Numerator $2(x-3)(x+2)$, denominator $(x-3)(x+3)$; $x\neq 3$ and $x\neq -3$ (both values needed).
A(ii) [1] The factor $x-3$ cancels, so there is a hole at $x=3$. For $x\neq 3$, $r(x)=\dfrac{2(x+2)}{x+3}$; at $x=3$ this is $\dfrac{10}{6}=\dfrac{5}{3}$. Hole at $\left(3,\dfrac{5}{3}\right)$.
B(i) [1] Only $x=-2$.
B(ii) [1] No. $x=3$ is not in the domain of r, so $r(3)$ is undefined; the cancelled factor gives a hole, not a zero.
C(i) [1] $x=-3$.
C(ii) [1] Near $x=-3$, $r(x)=\dfrac{2(x+2)}{x+3}$ with numerator close to -2. From the left the denominator is a small negative number, so $\lim\limits_{x\to-3^-} r(x)=\infty$; from the right it is a small positive number, so $\lim\limits_{x\to-3^+} r(x)=-\infty$ (both needed).
D(i) [1] $\lim\limits_{x\to\infty} r(x)=2$.
D(ii) [1] $y=2$. The numerator and denominator both have degree 2, so the end behavior is the ratio of leading coefficients $\dfrac{2}{1}=2$ in both directions, and $\lim\limits_{x\to-\infty} r(x)=2$ as well.

4. MARK MAP BY CLUSTER

Cluster	Lessons	Marks	Where
C1 Rates of change and concavity	1.1, 1.2, 1.3, 1.4	15	Q1 (6), Q2 (3), Q3 (3), Q5 (3)
C2 Polynomial zeros and end behavior	1.5, 1.6	6	Q4 (2), Q6 (4)
C3 Rational functions	1.7, 1.8, 1.9, 1.10	8	Q7 (8): D 1.7, B 1.8, C 1.9, A 1.10, two marks each
Total		29	Part A 14, Part B 15

5. READINESS BANDS AND WHEN AN INTERVENTION PLAN IS REQUIRED

Individual readiness (total out of 29).

Band	Marks	Action
Ready	22 to 29 ($\geq 75\%$)	Continue with the pacing. Practise the flagged cluster, if any, in the weekly mixed-review session.
Developing	18 to 21 (62% to 74%)	Continue, and add a 20-minute targeted clinic on each cluster scoring under 60%.
Intervention	0 to 17 ($< 60\%$)	Individual intervention plan: reteach the weakest cluster, rework the missed parts from Lessons 1.1 to 1.10, then a short retest.

Cluster flag. A cluster is flagged for a student when the student earns under 60% of its marks (C1 under 9, C2 under 4, C3 under 5), whatever the total.

Group decision. Count the students in the Intervention band and the class average for each cluster.

If	Then
30% or more of the group is in the Intervention band, *or* any cluster class average is under 70%, *or* two or more students are flagged on the same cluster	**An intervention plan is required.** Reteach the weakest cluster to the whole group before Lesson 1.11 (one full session: Diagnose, Targeted Instruction, Practice, Mastery Gate), then give a parallel retest of that cluster.
Fewer than 30% in Intervention, every cluster average at 70% or more, and no cluster flagged by two students	**No group intervention.** Individual clinics only for the students flagged above; carry on with Lesson 1.11.

Reading a weak cluster.

C1 low	Check Q1 B(i) and Q2 B(i) first: if average rate of change is wrong the rest follows. If the calculation is right but B(iii) is missed, the gap is concavity and the secant position, not arithmetic.
C2 low	Q4 B(i) missed all three zeros means calculator graphing and intersection skills; Q6 A and D missed means conjugate pairs and multiplicity.
C3 low	Q7 A and B missed means factoring and cancelling; C missed means one-sided limits and the sign of the denominator; D missed means degrees and leading coefficients.

A spreadsheet that applies these rules automatically is supplied with this exam: enter each student's mark for every part and it returns the total, band, flagged clusters and the group decision.